

Progress Report

April 18, 2026

DEVELOPER	ROLE	SPRINT PERIOD
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Summary

The April 18 sprint focused on API integration, environment configuration, and source control. The primary goal was to connect the ASP.NET Core MVC frontend to the backend API built by a teammate. This involved resolving a series of dependency injection errors, network connectivity issues, and database connection problems encountered while attempting to run both projects simultaneously. After exhausting network-based approaches, the backend was cloned locally and PostgreSQL setup was initiated. The session concluded with the new frontend codebase being pushed to the Artajo branch on GitHub with a clean, properly structured commit.

Tasks Completed

Resolved namespace and folder structure errors — fixed mismatched folder names (Service vs Services, Models) causing build failures	DONE
Registered ApiService and HttpContextAccessor in Program.cs — fixed dependency injection error on AccountController activation	DONE
Debugged backend connectivity — diagnosed and attempted to resolve SocketException errors when connecting to teammate's API over LAN	DONE
Investigated firewall and network access issues — ran netsh firewall rules, tested IP-based connections, and identified network permission blocks	DONE
Cloned backend repository locally — pulled the full project from GitHub and ran dotnet restore to resolve missing NuGet assets	DONE
Diagnosed PostgreSQL connection failure — identified missing local database instance as the cause of backend startup crash	DONE
Replaced old frontend files in Artajo branch — removed outdated implementation and pushed new ASM MVC project structure	DONE
Pushed to GitHub with proper commit message — used refactor commit type with descriptive body following Git best practices	DONE

Challenges Encountered

LAN connectivity and firewall restrictions

Multiple connection attempts to the teammate's API over the local network failed due to Windows Firewall blocking inbound and outbound connections on port 5167. Despite adding firewall rules and modifying `launchSettings.json` to bind on 0.0.0.0, access remained denied from the client machine.

Missing local PostgreSQL instance

After cloning the backend locally, the application crashed on startup because no PostgreSQL instance was running on the local machine. The backend depends on a live database connection to seed admin data, making it impossible to start without the database.

Git repository navigation complexity

Working within a nested cloned repository structure caused confusion with file paths and git context. Extra care was needed to ensure commands were run in the correct directory to avoid staging or committing to the wrong location.

04 Plans for Next Session

- Install and configure PostgreSQL locally — set up the database, run migrations using `dotnet ef database update`, and verify the backend starts successfully.
- Run both frontend and backend simultaneously — confirm API calls from the MVC frontend reach the local backend and return valid responses.
- Test the login flow end-to-end — verify JWT token is received, stored in session, and attached to subsequent API requests correctly.
- Begin testing remaining API endpoints — Students, Teachers, Courses, and Attendance pages to confirm data loads and displays as expected.